

Benign And Malignant Conditions Of The Breast

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OBJECTIVES

At the end of this presentation the student will be able to:

- Differentiate between benign and malignant breast conditions
- Explain the management of benign condition ,malignancy and infection of breast
- Explain the breast self Examination

Benign conditions of the breast

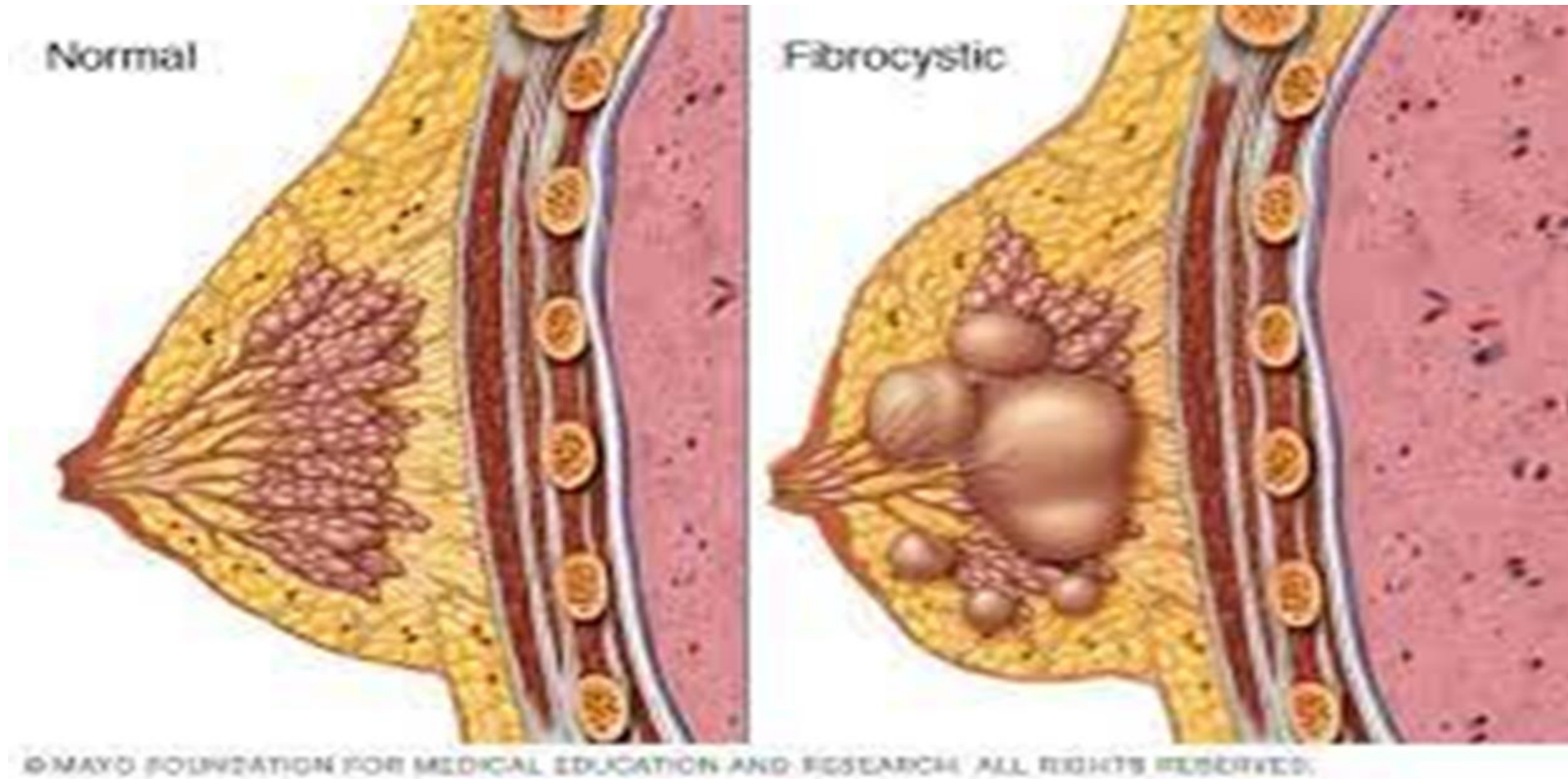
- A benign condition of the breast refers to a lump, cyst, or nipple discharge (fluid) of the female or male breast that is **not cancerous**.
- For women, the most common benign conditions are:
 - **Fibrocystic breast changes**
 - **Fibroadenoma**
 - **Phylloid tumor**
 - **Fat necrosis**
 - **Nipple discharge**
 - ***Intraductal papilloma***

Benign conditions...

❖ Fibrocystic breast changes

- Fibrosis feels like scar tissue, thick tissue, or a fluid-filled cyst and can be rubbery and firm
- Breasts will feel lumpy with Tender breast nodule
- These are usually small and common breast lumps in younger women
- determined by estrogen predominance and progesterone deficiency that result in hyperproliferation of connective tissue (fibrosis)
- This can happen to anyone but usually happens during breastfeeding
- Patients may be recommend to take birth control pills to reduce fluid buildup

Benign conditions ...

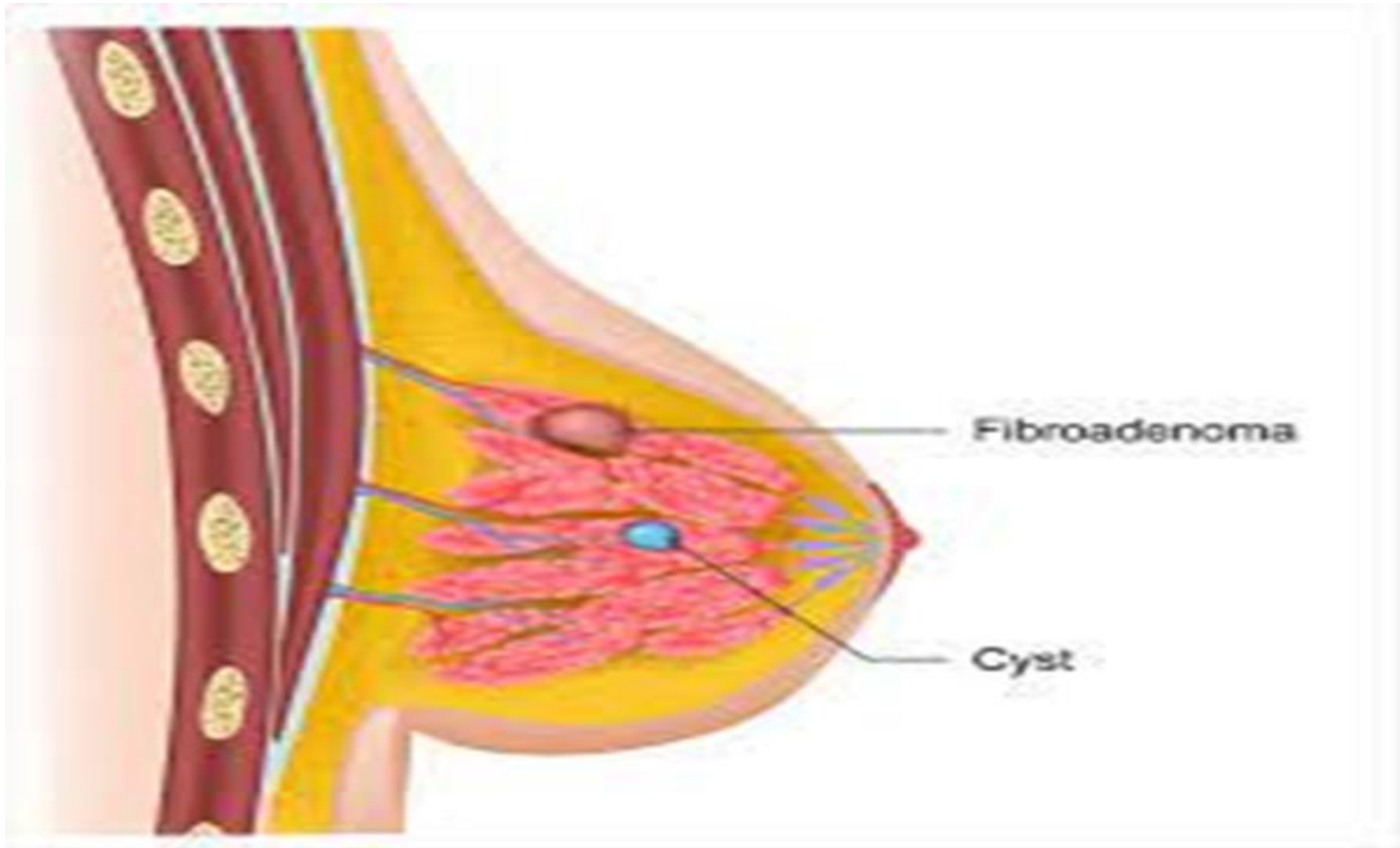


Benign conditions ...

❖ *Fibroadenoma*

- This represents a focal developmental abnormality of a breast **lobule**
- **Non-tender**
- **Feel like a small, Firm, round, moving marbles in breast.**
- Happen in premenopausal
- Tumor composed of **fibrous tissue and glands**
- As it is related to the use of birth control pills, an alternative birth control method is recommended. If the condition is painful, surgical removal of lump

Benign conditions ...



Benign conditions ...

❖ *Phyllodes Tumors*

- They are non-tender tumors that start in the **connective (stromal) tissue** of the breast
- Are benign and only a small number are malignant (cancer).
- Phyllodes tumors account or less than 1% of breast neoplasms
- Common in **postmenopausal** women
- Local therapy include **surgery** to remove the tumor and extra tissue around it.
- Sometimes, radiation is given as well
- Systemic therapy (chemotherapy) may be used if a malignant phyllodes tumor has spread outside of the breast, though this not common.

Benign conditions ...

❖ Fat necrosis

- lumps formed when areas of **fatty breast tissue are damaged**
- Is a side effects of breast **surgery, radiation, other trauma**
- A lump that may feel round and hard.
- It's common in **extremely overweight women**
- It may be filled with fluid fat
- Usually, fat necrosis doesn't require further treatment.
- If it contains fluid (oil cyst), drain the fluid from the cyst with **a fine needle aspiration**



Benign conditions ...

❖ Nipple discharge

- Pathologic nipple discharge is a spontaneous single-duct **discharge that is serous or bloody**
- **Multiduct discharges** that are elicited only following manual expression are physiologic
- Spontaneous milky nipple discharge, also called **galactorrhea**
- Most pathologic nipple discharges are caused by benign **intraductal papillomas**
- Underlying malignancy ranges from 2% young to 20 % or older women

Benign conditions ...

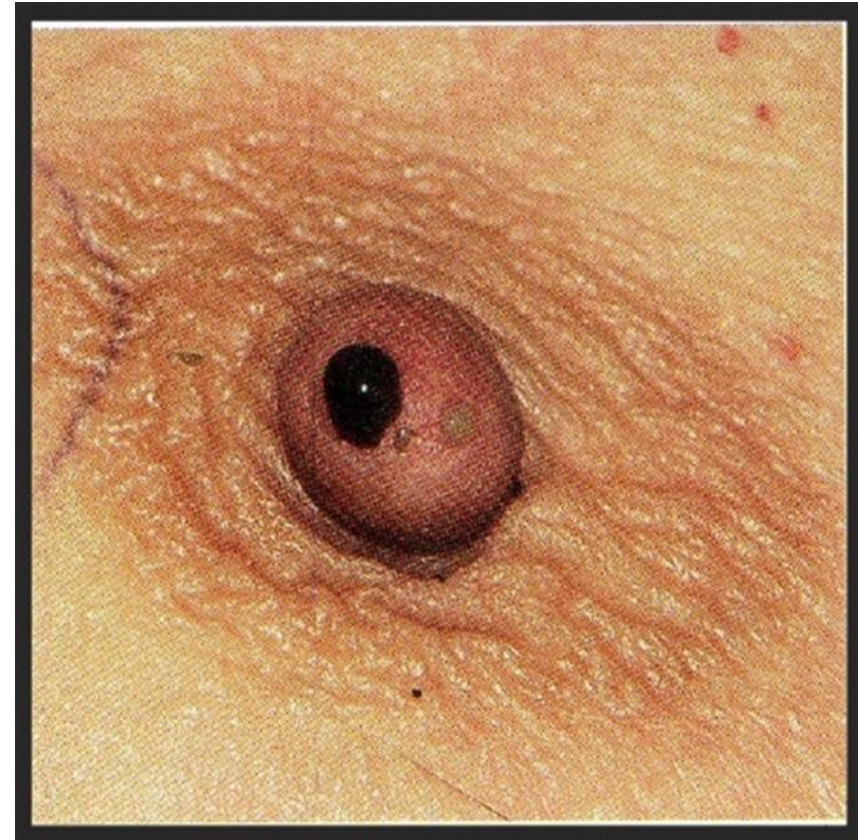
- The fluid coming from nipple may be different colors.
- Clear or milky color represents a problem with hormones
- If the discharge is green-black, it represent a blocked milk duct
- If the discharge is bloody, it could be related to an injury, infection, or benign tumor. It also can be associated with breast cancer
- Treatment depends on the underlying cause of nipple discharge (lump, infection, or cancer).

Benign conditions ...

Single duct milky discharge



Bloody discharge



Benign conditions ...

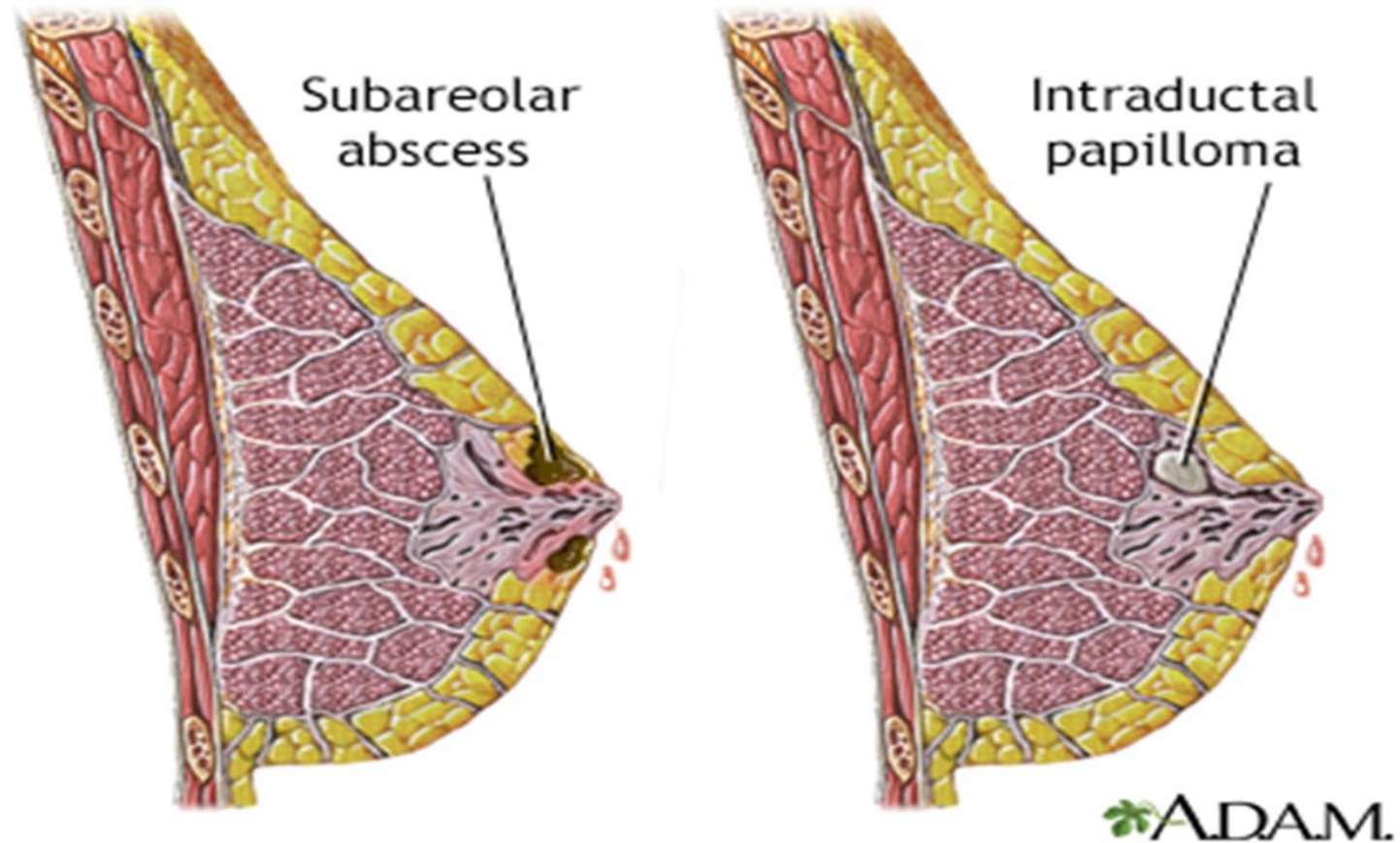
Less common types of benign breast conditions include:

❖ *Intraductal papilloma*; Small tumors can form in nipple's milk ducts.

- ➔ Papilla grows inside the duct
- ➔ Papilla: is a finger like outgrowth lined by epithelium
- ➔ Bloody nipple discharge
- ➔ Premenopausal
- ➔ Treatment is Based on pain and discomfort. surgically removal is considered.

Benign conditions ...

Common causes of abnormal nipple discharge



Risk factor of benign and malignant breast conditions

- Age >50
- Family History of breast CA
- Prior breast cancer
- Hormone therapy(estrogen)
- Birth control pills
- Infection (mastitis)
- Pregnancy& Breastfeeding
- Early menarche
- late menopause
- null parity
- older age at first pregnancy
- Radiation exposure
- Obesity
- Alcohols &Smoking

Common signs and symptoms

- ↘ Breast tenderness
- ↘ Breast Pain (mastalgia)
- ↘ swelling in the breast
- ↘ A lump that can be felt through the skin or nipple
- ↘ Change in breast size, shape or appearance
- ↘ Skin irritation

Common signs and symptoms...

- ↘ Redness on the nipple and skin of the breast
- ↘ Pain in the nipple
- ↘ Retraction of the nipple (part of the nipple looks like it is pulling inward)
- ↘ Abnormal discharge from the breast (that is not milk)
- ↘ Dimpled, puckered or scaly breasts

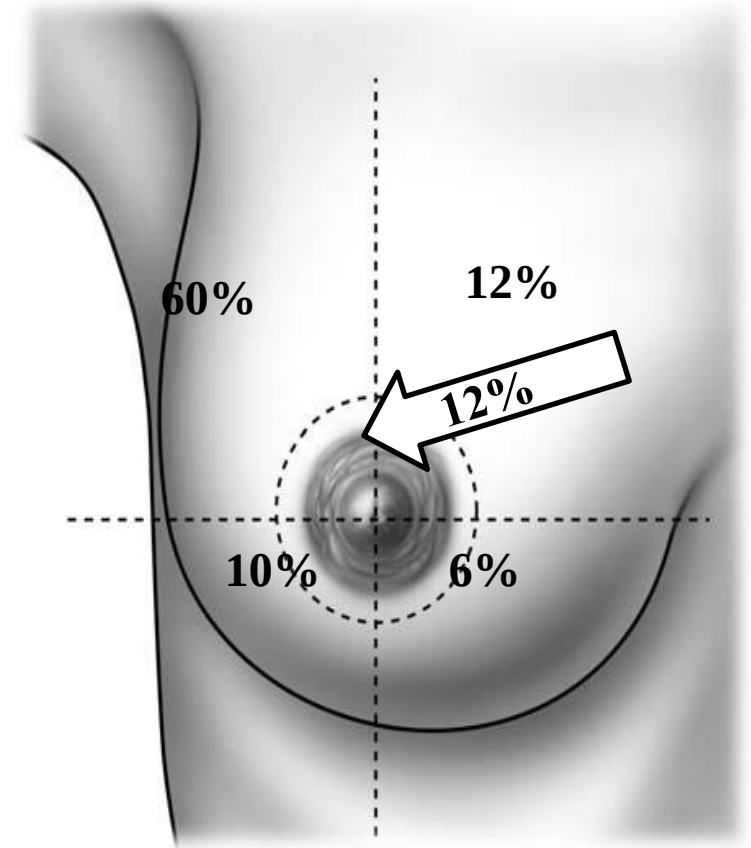
Malignant conditions of the breast

- Malignant breast disease refers to harmful, cancerous abnormalities or growth in the breast tissue
- It occurs when cancerous cells begin to **mutate and spread throughout the healthy tissue**
- Is the **most common** cancer in **female(25%)** worldwide and **31.5%** in Ethiopia
- **The second** common cause of cancer death in women; 13 % (WHO 2020)
- Main cause of death in women ages **40-59 years**
- **Invasive ductal carcinoma (IDC):** is the commonest type

Relative distributions Breast cancer

❖ To describe clinical findings the breast is often divided into **four quadrants**

- Upper outer quadrant ~ **60%**
- Upper inner quadrant ~ **12%**
- Lower outer quadrant ~ **10%**
- Lower inner quadrant ~ **6%**
- Central quadrant ~ **12%**



Features of Breast cancer

- All type of cancer, irrespective of their origin, occur due to this **disturbed growth of cells**
- Have longer life spans than the normal once
- Can also invade other tissue (metastasis)
- They grow into tumors that are supplied by an new network of blood vessels
- The ethology of breast CA is still poorly understood with known risk factors

Classification of Breast cancer

❖ **Carcinoma in situ, CIS:** are neoplasms, histologically corresponding to invasive carcinomas, but **without penetration of the basement epithelial membrane**

❖ There are **two types** of carcinoma in situ lesions in breast

1. Ductal carcinoma in situ(DIS); abnormal cells start in the cells **lining the ducts** without growing (invading) through the walls of the ducts into the tissue of the breast

- Also called stage 0 breast CA
- 20% of all breast CA and bilateral in 10-20%
- Anatomic precursor of invasive ductal carcinoma

Classification of breast cancer ...

2. Lobular Carcinoma in situ(LIS): arises in ductolobular units of breast parenchyma,

- They are bilateral in 1/3
- They have much smaller malignant potential than DCIS, and are not considered a true carcinoma
- Marker of increased risk for invasive breast carcinoma, not anatomic precursor

Classification of breast cancer ...

❖ **Invasive carcinoma:** Invasive carcinoma is the most frequent malignancy in women

1. Invasive (infiltrating) ductal carcinoma (IDC): It starts in the lining of milk ducts

- the most common type ~ **70- 80%**,

2. Invasive (infiltrating) lobular carcinoma (ILC): This cancer starts in the cells lining the milk glands (the lobules)

- the 2nd most common type ~ **5 – 10%**

Classification of breast cancer ...

3. Inflammatory breast cancer (IBC): This is a rare type of invasive breast cancer

- makes the skin of the breast look red and feel warm
- It also may make the skin look thick and pitted, something like an orange peel

4. Adenocarcinoma:- A type of carcinoma that starts in gland tissue

- Nearly all breast cancers are adenocarcinomas

5. Sarcoma:- A cancer that starts from connective tissues such as muscle tissue, fat tissue or blood vessels

Spread of breast cancer

■ Local spread with in the breast

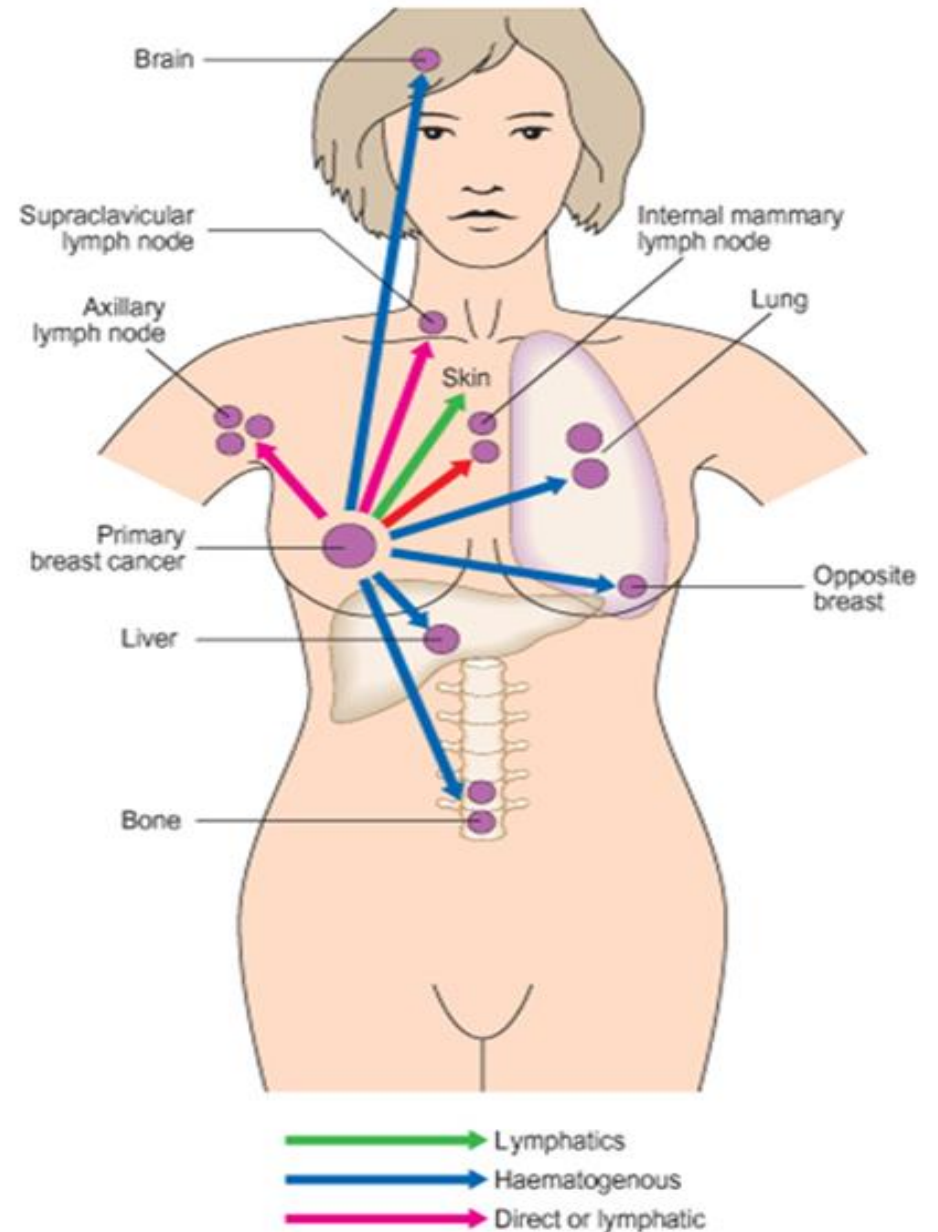
- involves the skin & fascia
- chest wall and other portions of the breast

➤ Regional spread of breast cancer

- axillary LNs
- internal mammary and Supraclavicular LN

➤ Hematogenous (distant) spread

- bone, lung, pleura, soft tissues, and liver.



Diagnosis of benign and malignant breast conditions

❖ History

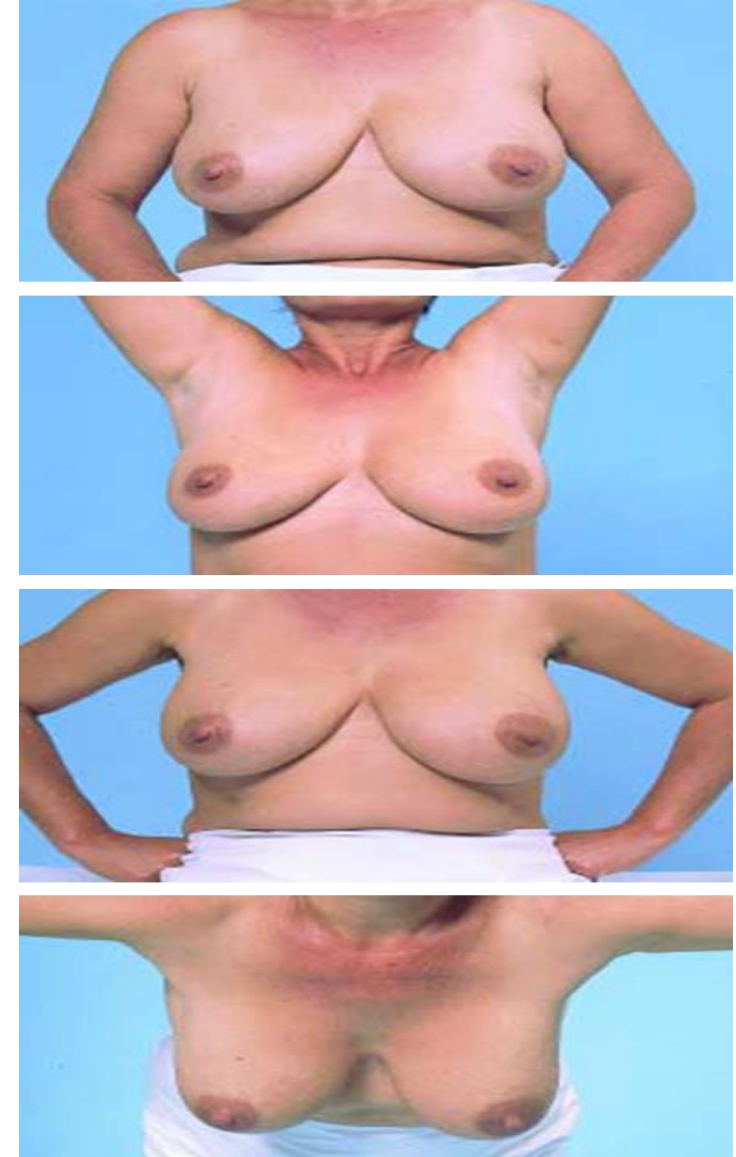
- age at menarche and menstruation status (pre or postmenopausal)
- age at first delivery
- family history of breast cancer
- duration of illness
- associated symptoms (pain, headache, cough, nipple discharge)

Diagnosis of benign and malignant breast conditions...

❖ Physical examination

➤ **Inspection:** symmetry, size, and shape, nipple or skin retraction or erythema

❖ arms extended forward in a sitting position leaning forward to accentuate any skin retraction



Diagnosis of benign and malignant breast conditions...

- **Palpation:** location, size, consistency, shape, mobility, fixation
 - ✓ examine all quadrants
 - ✓ avoiding a grasping or pinching motion
 - ✓ assesses all three levels of axillary lymphadenopathy



Diagnosis of benign and malignant breast conditions...

❖ Investigations

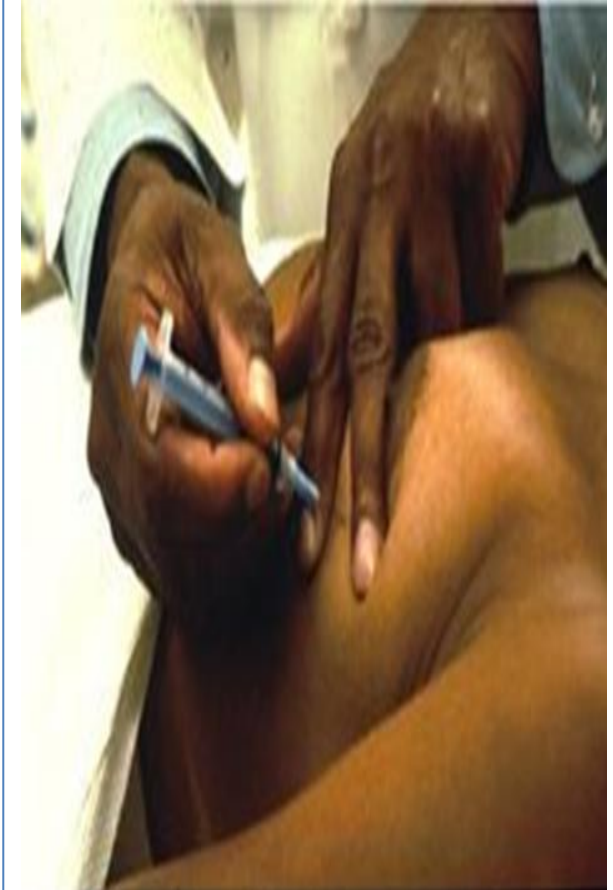
❖ Imaging tests:

- It includes Ultrasound scans, MRI, Mammography, ductography
- low-dose x-rays of the breast, CXR, CT

❖ Laboratory analysis:

- examines the presence of cells the nipple discharge under a microscope
- CBC, Blood group & Rh, FNAC & LFT

❖ Biopsy: Core needle biopsy,



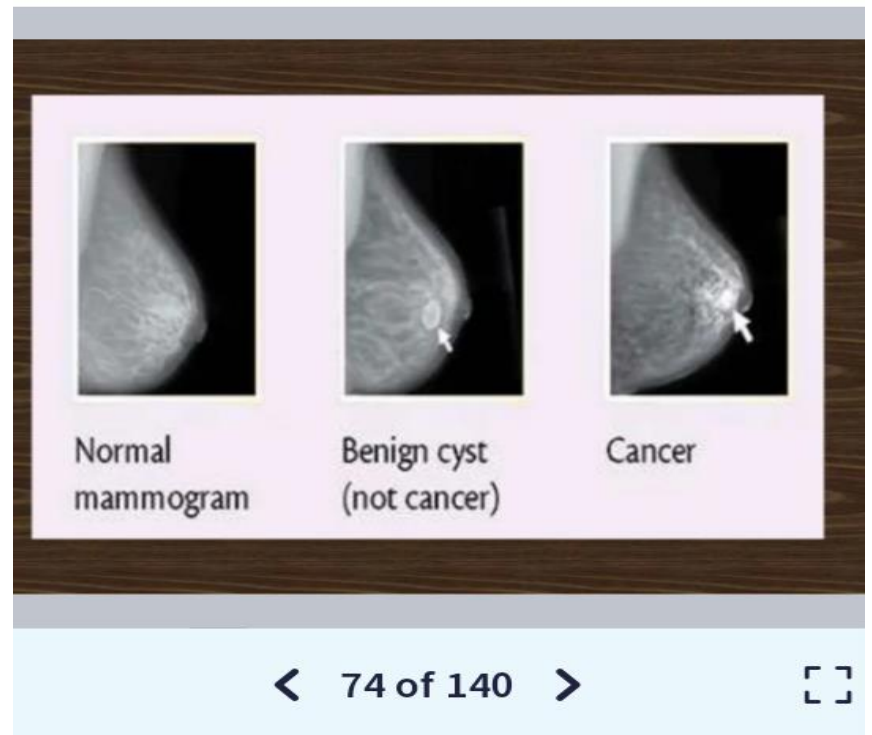
Diagnosis of benign and malignant breast conditions...

✓ Characteristics of malignant lesions in mammography:

- architectural distortion
- solid mass with or without stellate features
- micro calcifications
- stippled calcifications
- asymmetric thickening of breast tissues
- nipple retraction



Download Now



What is breast cancer
Types of breast cancer

Diagnosis of benign and malignant breast conditions...

❖ Triple Assessment

- Any patient with a breast lump or other symptoms suspicious of carcinoma, the Diagnosis should be made by a *combination* of:
 - 1. Clinical assessment*
 - 2. Radiological imaging and*
 - 3. Tissue sample (cytological or histological)*
- Positive predictive value is 99.9%
- If any of the **three assessment** suggests malignancy, the lump should be excised regardless of results from the other two

Stages of breast cancer

Stage1A: the cancer is 2cm or smaller and has not spread out side the breast.

Stage1B: The cancer in the breast is 2cm and few cancer cells are found in the lymph nodes under the arm

Stage2A: the cancer in the breast is larger than 2cm but smaller than 5cm and no cancer found under arm

Stage2B: The cancer in the breast is larger than 2cm but smaller than 5cm. Cancer is found one to three lymph nodes under the arm or near to breast bone.

Stages of breast cancer...

Stage3A: The cancer in the breast measures up to 5cm and cancer is found 4-9 lymph nodes under the arm.

Stage3B: Means cancer in can be any size and has spread to the skin of the breast or chest wall.

Stage3C: means cancer in the breast can be any size, may have spread to the skin of the breast or chest wall and cancer is found in 10 or more lymph nodes under the arm

Stage4: the cancer can be any size, the lymph nodes may or may not contain cancer, but the cancer has spread to other part of the body, such as distant lymph nodes, lung,bones, liver or brain

Management of breast cancer

- ✓ The treatment of breast CA will depend on
 - ❖ Type of breast cancer
 - ❖ Stage and grade of the cancer
 - ❖ Whether or not the cancer cells are sensitive to hormones
 - ❖ The overall health of the patient
 - ❖ Preference of the patient

Management of breast cancer...

- Treatment for breast CA entails:
 - **Local control**
 - ✓ *surgery & radiotherapy*
 - **Systemic control**
 - ✓ *hormone & chemotherapy*

Management of breast cancer...

1. Treatment of early breast cancer (Stage I & II)

- Breast conservation - *resection* of the primary breast ca with a normal margin, adjuvant *radiation* therapy, and assessment of regional *lymph node status*
- *Mastectomy* with sentinel lymph node and/or axillary LN dissection
- *Hormonal therapy*: E.g: Tamoxifen therapy

2. Locally advanced breast cancer (Stage-III):Neoadjuvant *chemotherapy*

- Usually a *modified radical mastectomy* (MRM)
- Followed by adjuvant *radiation* therapy

Management of breast cancer...

3. Breast CA with distant metastasis (Stage IV)

- provide palliation
- symptomatic relief

❖ Treatment

- Combination chemotherapy
- Total mastectomy
- Radiotherapy
- Tamoxifen therapy in ER positive

Complications of mastectomy

- **Seromas** - the most common
 - Build up of clear bodily fluids in a place on the body where tissue has been removed by surgery
- **Wound infections**
- **Hemorrhage**
- **Lymphedema** - increased risk in:
 - extensive ALND
 - the delivery of radiation therapy
 - the presence of pathologic lymph nodes & obesity
- **Nerve injury**

Breast self examination (BSE)

- **BSE** Detects the majority of breast abnormalities
- The best time to examine breasts is usually **1 week after menstrual period starts**, when breasts are least likely to be swollen or tender
- If menstrual cycle is irregular, or if menstruation stopped due to menopause do examination on a **day of the month that's easy to remember**

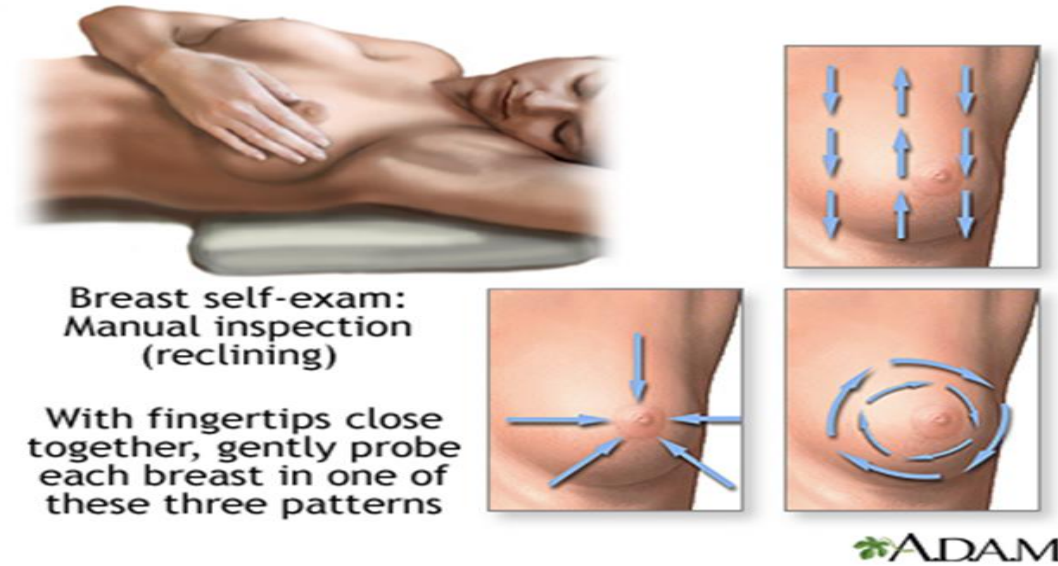


Breast self examination (BSE)...

- To do a breast self-examination
 - Lie down or Stand upright in front of mirror
 - you may also check your breast while taking shower

❖ Inspect for:

- Skin changes
- Redness
- Visible bumps
- Nipple crusting
- Symmetry

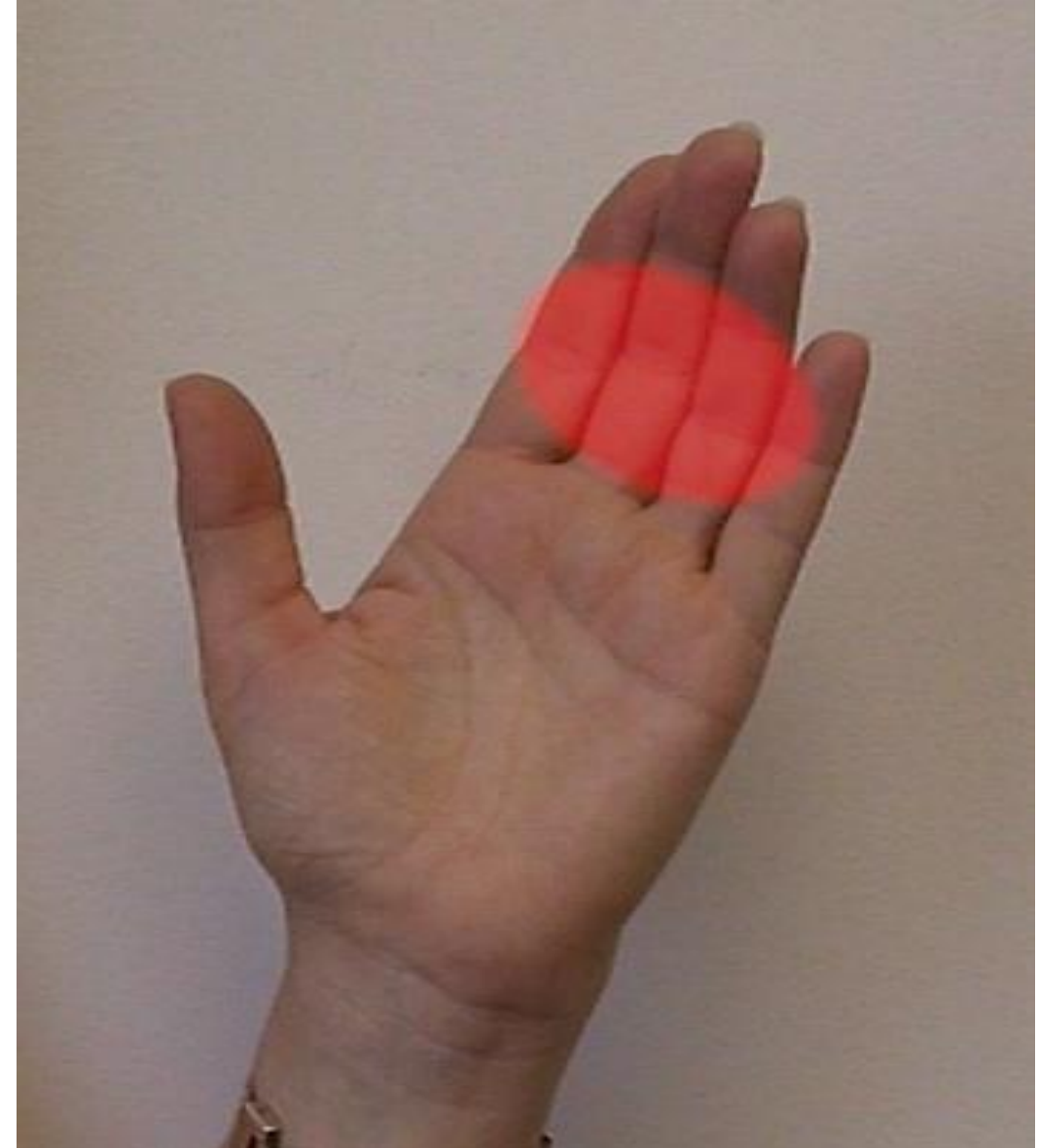


Breast self examination (BSE)...

- ✓ Check the entire breast by feeling all of the tissue from the collarbone to the bottom of the bra line and from the armpit to the breastbone
- ✓ Use the middle fingers of your left hand to check your right breast and vice versa
- ✓ Use three different levels of pressure (Light, Medium , and firm) to feel all of your breast tissue
- ✓ If you find the same kind of lump in the same area on the other breast, both breasts are probably normal

Breast self examination (BSE)...

- **Use the Middle of our Fingers**
 - Fingertips are too sensitive (all breasts are somewhat lumpy)
 - Palm is too insensitive
 - **Middle portion of fingers is just right**



Breast self examination (BSE)...



➤ **Raise Arms Up**

- Breasts should rise evenly
- Watch for dimpling or retraction

Breast self examination (BSE)...

➤ **Feel for Lumps**

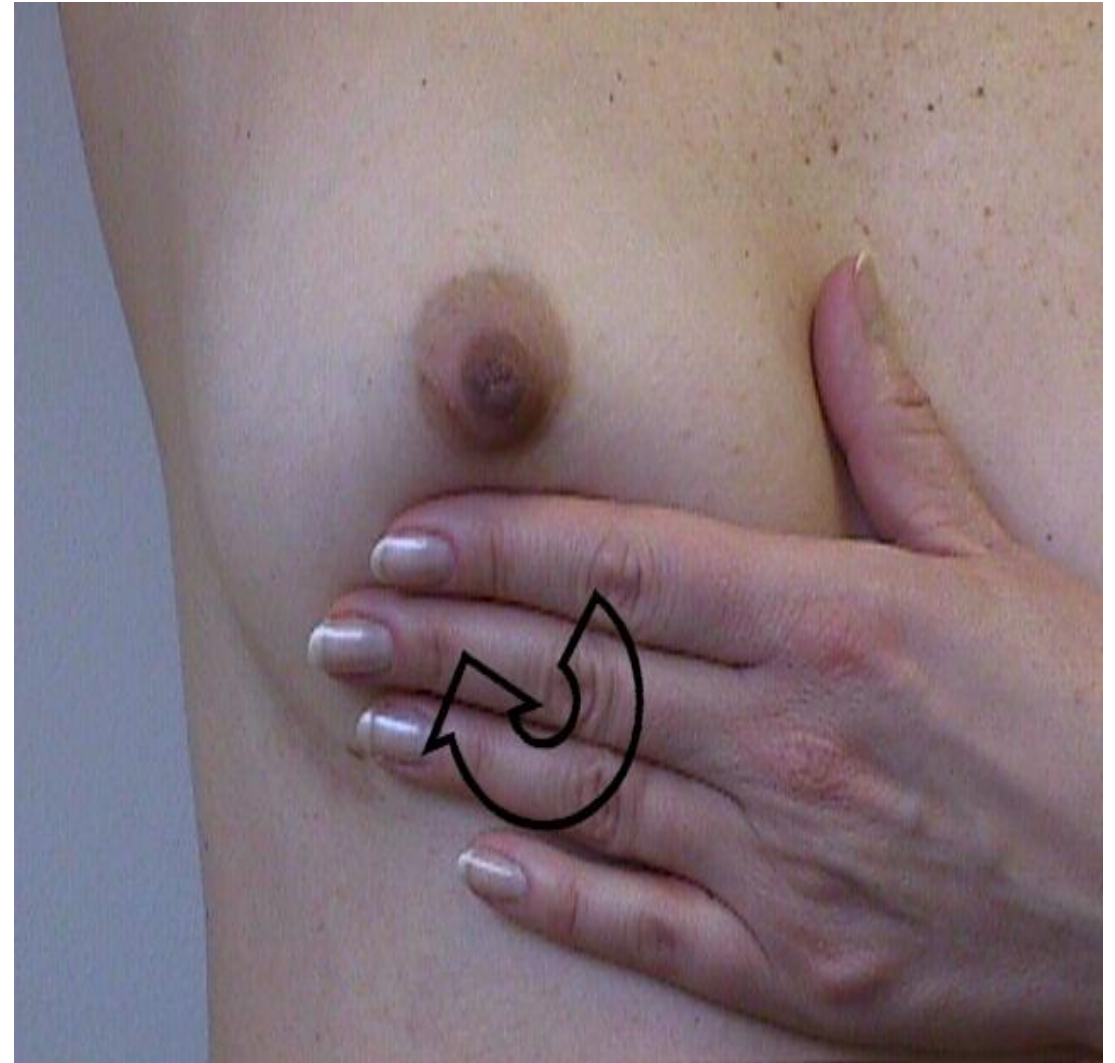
- Raise the arm
- Feel with opposite hand
- Feel for a “marble in a bag of rice”



Breast self examination (BSE)...

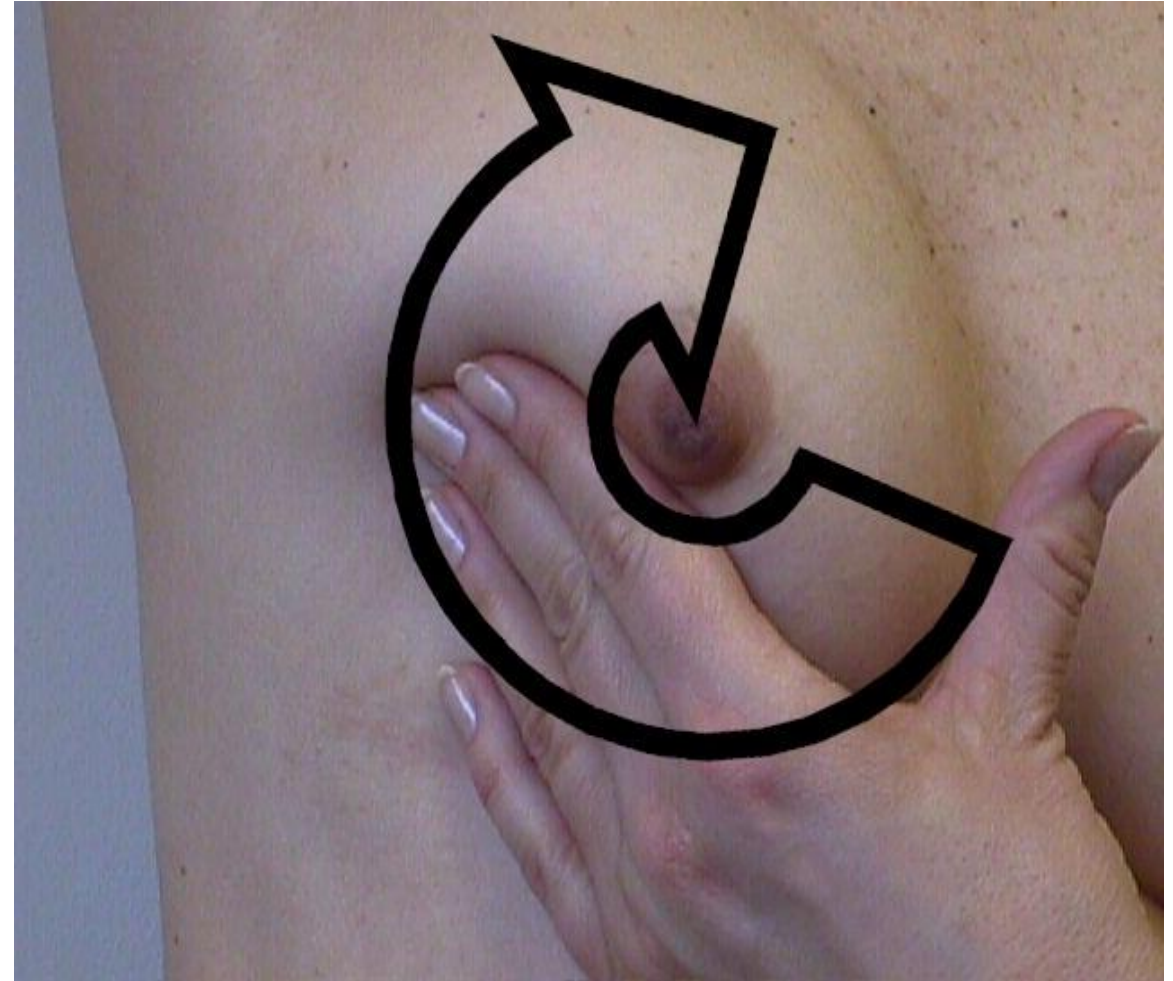
➤ Move hand in small circles

- Stay in one place
- Press in while circling with your hand
- Feel for thickenings the size of a marble



Breast self examination (BSE)...

- **Then move to another location**
 - Work your way around the breast in a **clockwise fashion**, using small circles of the hand as you go
 - Make sure the entire breast is felt



Breast self examination (BSE)...

➤ The “Tail” of the Breast

- Breast is not perfectly round.
- A “Tail” of breast tissue normally extends into the armpit
- Make sure to feel for lumps in that portion of the breast



Breast self examination (BSE)...



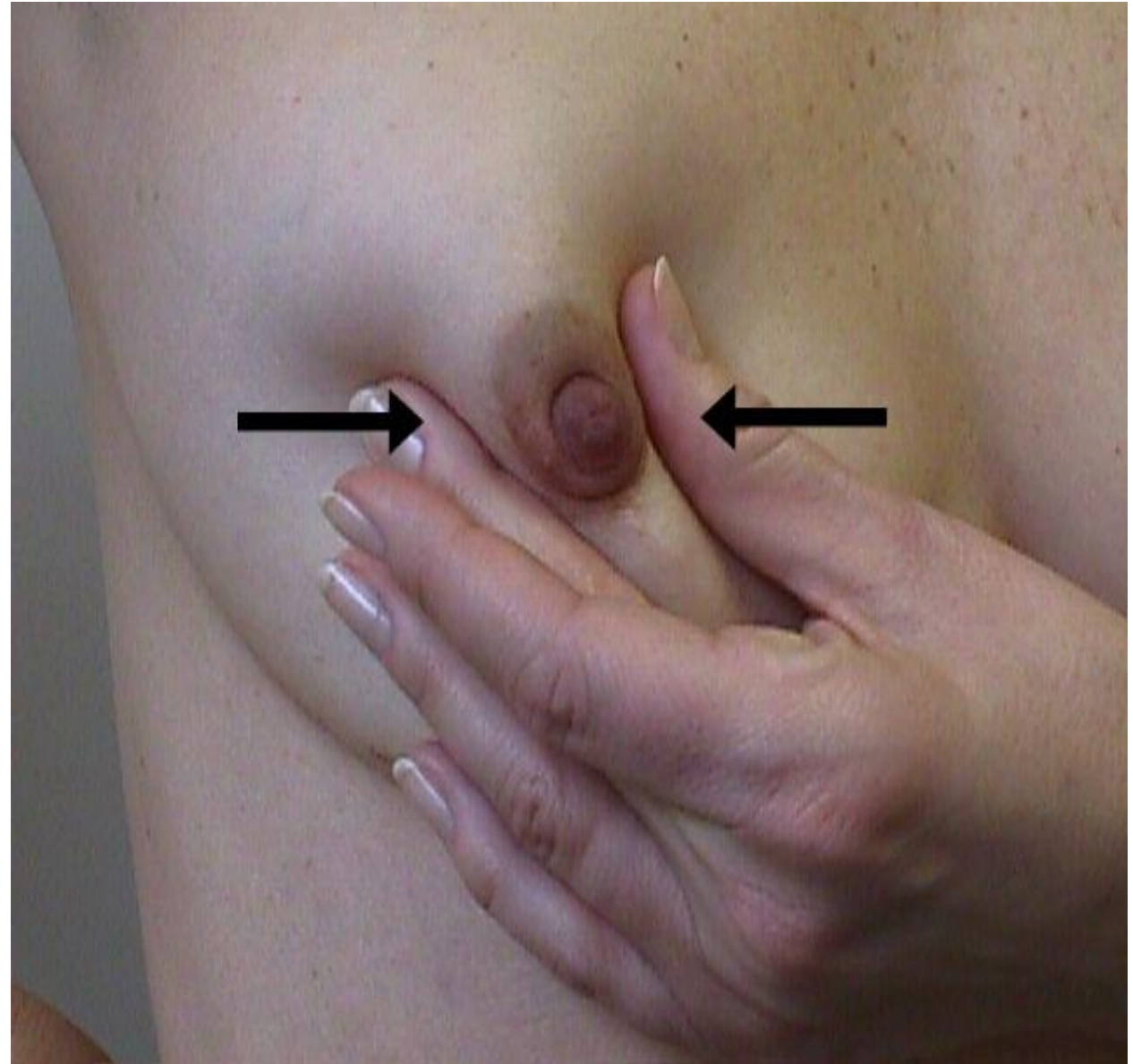
➤ Feel the Armpit

- Use the same circular motions.
- Feel for breast lumps and lymph nodes.
- Normal lymph nodes cannot be felt.
- Enlarged lymph nodes are about the size of a pencil eraser, but longer and thinner.

Breast self examination (BSE)...

➤ Try to Express Nipple Discharge

- Strip the ducts towards the nipple
- Normally, one or two drops of clear, milky or green-tinged secretions
- Should not be **bloody or in large quantity, squirting out or staining the inside of a bra.**



Breast self examination (BSE)...

➤ Now Check the Other Side

- Follow same maneuvers
- Raise the arm above your head
- Feel for lumps or masses



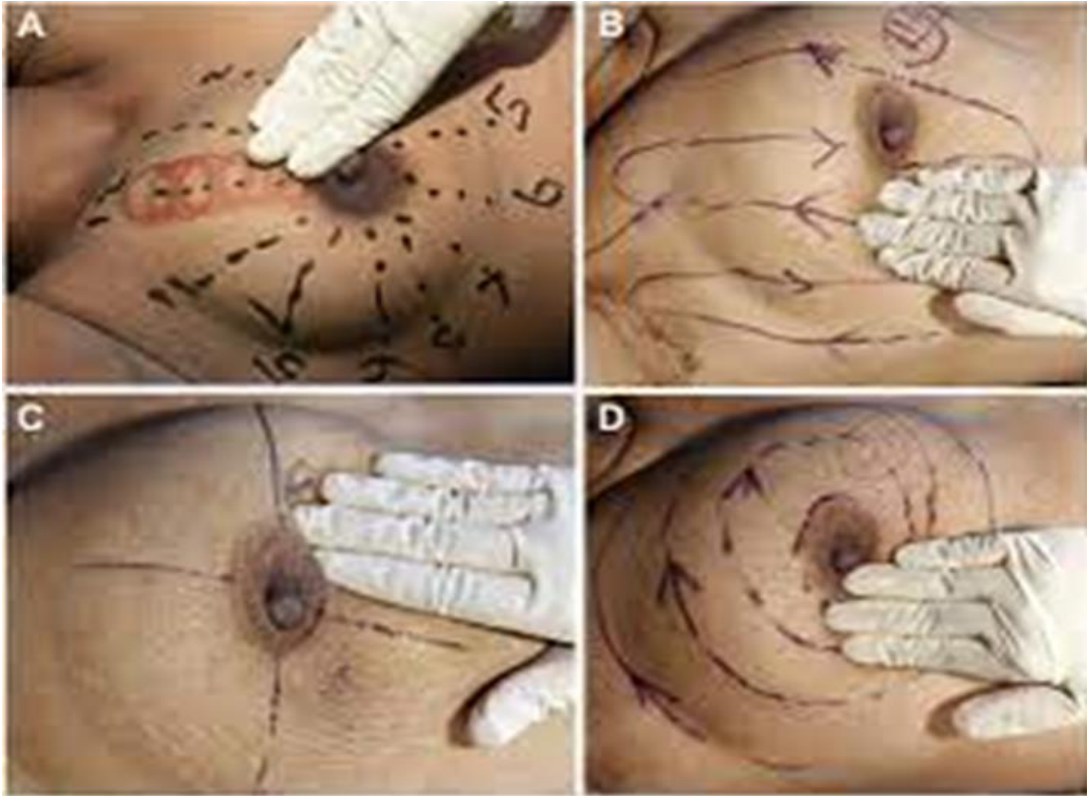
Breast self examination (BSE)...



- Don't be panic if you find something in your breast
- Most (90%) of breast lumps are benign
- You must be seen by physician or other qualified health care provider right away.

Have a Professional Breast Exam Each Year

- Basically the same maneuvers
- Ask questions



Benign & Malignant Disease of the Cervix

Cervical Intraepithelial neoplasia (CIN)

- CIN means **disordered growth and development of the epithelial lining** of the cervix
- CIN is used to describe histologic findings of **cervical biopsy**
- **CIN is most commonly detected in women in their 20s**
- **Nearly all cervical neoplasia**, both squamous and columnar, develops within the **transformation zone**, usually adjacent to the new SCJ

Cervical Intraepithelial Neoplasia (CIN)...

➤ **CIN has three degrees of severity:**

- **CIN 1:** Mild atypical cellular changes in the **lower third of the epithelium** (formerly called **mild dysplasia**).
- **CIN 2:** Moderate atypical cellular changes confined to the **basal two-thirds of the epithelium** (formerly called **moderate dysplasia**)
- **CIN 3:** Severe atypical cellular changes encompassing **greater than two-thirds of the epithelial thickness, and includes full-thickness lesions** (formerly called **severe dysplasia or carcinoma in situ**).

Cont...

Normal cervix



Cervical dysplasia



Normal cervical cells



Cancerous or pre-cancerous cervical cells



Risk Factors for Cervical Neoplasia

- Infrequent or absent cancer screening Pap tests
- Early sexual intercourse
- Multiple sexual partners
- Male partner who has had multiple sexual partners
- Cigarette smoking
- Family history
- **Cervical high-risk human papillomavirus infection**
- nullparity
- Immunosuppression (**HIV**)

Human papilloma virus (HPV)

- HPV is a nonenveloped DNA virus with a protein capsid
- It infects epithelial cells exclusively and approximately 30 to 40 HPV types have an affinity for infecting the lower anogenital tract
- HPV is a nonlytic virus, and therefore infectiousness depends upon desquamation of infected cells
- More than **100 HPV** types have now been identified.

Human papilloma virus (HPV)...

- Clinically, HPV types are classified as high-risk (HR) or low-risk (LR) based upon their cervical cancer oncogenicity.
- Low-risk HPV types **6 and 11** cause nearly all **genital warts**
- In contrast, the high-risk- HPV types include 16, 18, 31, 33, 35, 45, and 58 and account for approximately 95% of cervical cancer cases worldwide
- However, **HPV 16 & 18** are the dominant causes of cervical cancer
- **HPV 16** is the dominant cancer-related HPV, accounting for 40 to 70% of invasive squamous cell cervical cancers worldwide

Human papilloma virus (HPV)...

➤ **Transmission**

- **Sexually transmitted**
- **Vertical infection** (Conjunctival, laryngeal, vulvar, or perianal warts present at birth or develop within 1 to 3 years)

➤ **Diagnosis of Infection**

- Clinical lesions and results of cytology, histology, and colposcopy, all of which are **subjective and often inaccurate.**
- Definitive diagnosis is made by direct detection of HPV DNA

Management of CIN

Management of low grade lesions (CIN I)

- CIN 1 can be observed indefinitely, especially in adolescents, or
- Treated if it persists for at least 2 years

Management of High Grade lesions

- **CIN 2 adolescents same as CIN I**
- **CIN 2 in adult women and CIN 3 are treated by excision or ablation**

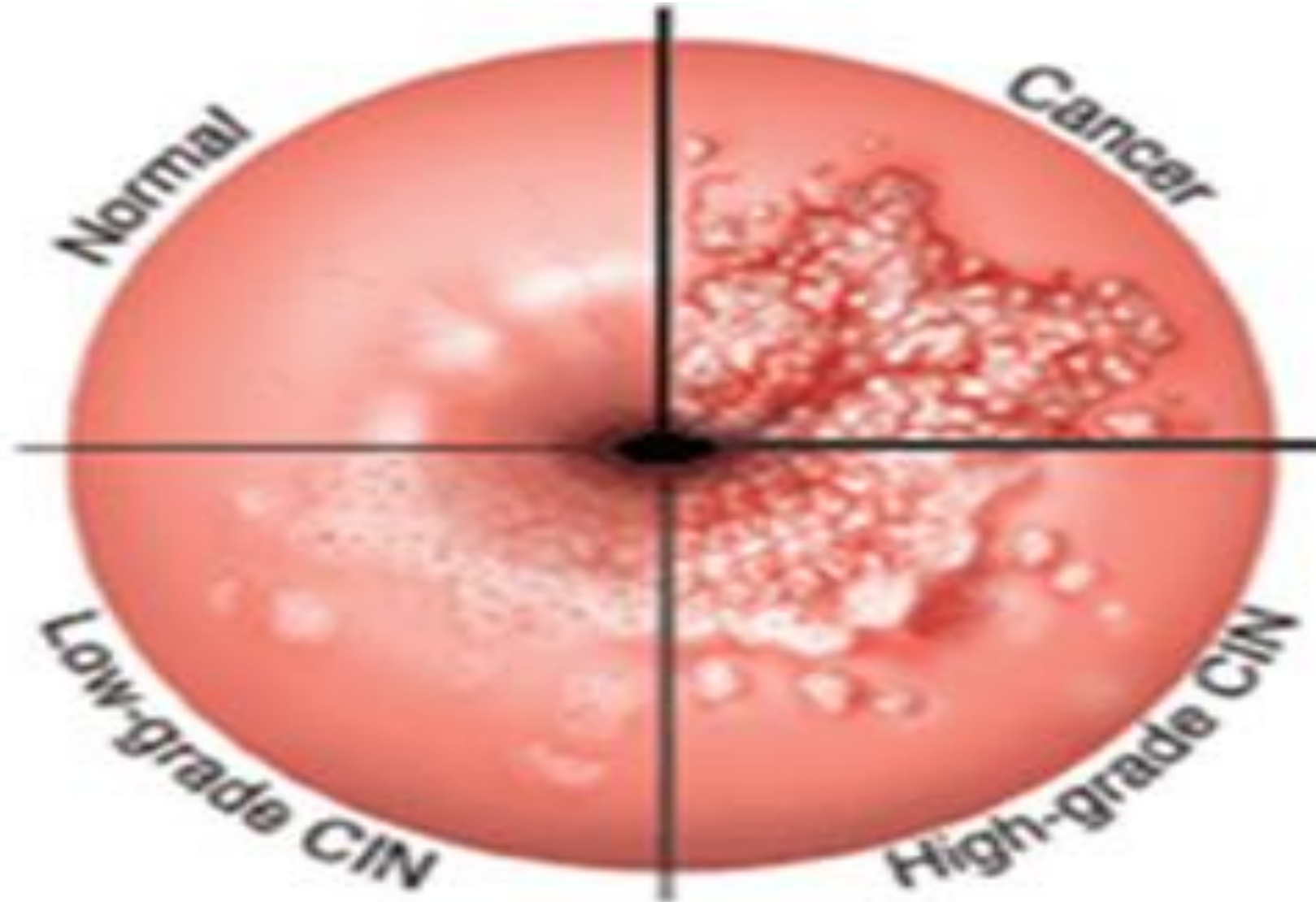
Cervical Cancer

Objectives

By the end of this lesson students could be able to:

- Define cervical cancer
- Describe the risk factors of cervical cancer
- Discuss the diagnosis of cervical cancer
- Describe the cervical cancer screening methods
- Identify the prevention methods of cervical cancer

Cont...



Introduction

- Cervical cancer is a malignant tumor of the lower most part of the uterus which connects it to the vagina
- Cervical cancer is the fourth most common cause of cancer death in the world and second most common cancer type among women in the developing countries
- Human Papillomavirus (HPV) infection is responsible for 99% of cervical cancer

The common Types of cervical Cancer

- Squamous cell Carcinomas
 - Cancer of flat epithelial cell
 - 80% to 90%
- Adenocarcinomas
 - Cancer arising from glandular epithelium
 - 10% - 20%
- Mixed carcinoma
 - Features both types

Signs and Symptoms

- Vaginal bleeding
- Menstrual bleeding is longer and heavier than usual
- Bleeding after menopause or increased vaginal discharge
- Bleeding following intercourse or pelvic exam
- Pain during intercourse
- Exophytic or endophytic growth
- Polypoid mass, papillary tissue, or barrel-shaped cervix
- Cervical ulceration or granular mass

Staging of Cervical Cancer

➤ Cervical cancer is one of the two gynecologic malignancies (vaginal cancer being the other) that still uses a clinically based staging system

Stage I: confined to the cervix

IA1 – <3mm depth of invasion

IA2 – stromal invasion 3-5mm in depth
or <7 mm in width

IB1- tumor < 4 cm

IB2 - tumor > 4 cm in diameter

Stage II: extension beyond cervix

IIA – upper 2/3 of vagina

IIB – Parametrium involvement

Cont...

Stage III: Tumor extends to pelvic sidewall

IIIA – lower 1/3 of vagina

IIIB – extension to pelvic sidewall or hydronephrosis

Stage IV: extension beyond pelvic area

IVA – bladder or rectal mucosa

IVB – distant metastases

Diagnosis

- ❖ **Cervical Cytology (Pap Test)**
- ❖ **Colposcopic examination**
- ❖ **Iodine-non staining (Schiller-positive)**
- ❖ **Endo cervical curettage**
- ❖ **Cone biopsy**

Treatment Modalities

- **Surgery and radiation** therapy are the two primary therapeutic modalities most commonly used to treat invasive cervical carcinoma.
- Primary surgical management is limited to patients with **stage I and IIA disease**,
- while radiation therapy can be applied to patients with all stages of disease
- **Concurrent Chemotherapy and Radiation Therapy**

Cont...

- **Cervical conization**
- **Simple hysterectomy**
- **Radical hysterectomy**
- **Radiation therapy with chemosensitization**
- **Palliative treatment**

Cervical cancer screening

- Cervical cancer screening is a method of detecting precancerous lesions in the cells of the cervix among women who have no symptoms
- It has been recognized to be a very effective prevention strategy for cervical
- Evidences showed that when women are screened **at least once in their lifetime** for cervical cancer, could decrease their risk of cancer **by 25–36%**

❖ Organized screening

Is designed to reach the highest possible number of women at greatest risk of cervical cancer with existing resources

❖ Opportunistic screening

Is screening done on women who are visiting health services for other reasons

Cervical cancer screening ...

➤ **Methods of cervical cancer screening include:**

1. Cervical Cytology

2. HPV DNA testing

3. Combined test

4. Visual method

a) Visual inspection with acetic acid (VIA)

b) Visual inspection with Lugol's iodine(VILI)

Prevention of cervical cancer

❖ Behavioral Interventions

❖ Condoms use

❖ **Having the Pap Test:** 3 years after first vaginal intercourse or by age 21, Have test annually

❖ Prophylactic Vaccines

- Recombinant quadrivalent vaccine against types 6, 11, 16, and 18 (Gardasil)
- Indicated for women aged 9 to 26 years
- Administered in three intramuscular doses over 6 months

THANK YOU!